

Theory

Gaussian Beam Optics

Tyler Celotta
UAF

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Abstract

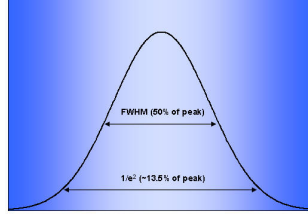
The purpose of the document is to work through the theory and experimental design for optimizing second-harmonic generation (SHG) at 1064 nm using a periodically poled lithium niobate (PPLN) crystal ($50 \times 2 \times 0.5 \text{ mm}^3$). In this document, I define Gaussian beam parameters, including beam waist, Rayleigh range, divergence, and confocal parameter, in order to determine the optimal focusing geometry. The Boyd-Kleinman condition for maximum CW SHG efficiency ($L/b = 2.84$) is a target focused waist of approximately $54.6 \text{ }\mu\text{m}$, but experimental constraints redefines a target of $77\text{--}80 \text{ }\mu\text{m}$ ($L/b \approx 1.3 - 1.4$). Starting from a collimated 0.56 mm beam at 1064 nm with a measured divergence of 1.21 mRad , the collimator waist is estimated to be $\omega_{01} \approx 280 \text{ }\mu\text{m}$, located 6.2 mm behind the collimator face. That being said, the calculated waist radius of roughly 0.279 mm would result in a beam larger than the crystal itself can accommodate. To account for this, I used an ABCD ray-transfer matrix to propagate a beam, constructing two equations relating a chosen lens focal length f to the required lens position d_1 and image distance d_2 , with magnification $\alpha \approx 0.286$.

1 Experiment Specifics

1. Wavelength: 1064 nm
2. Collimator output beam size: $0.56 \text{ mm } \frac{1}{e^2} \text{ diameter}$
3. PPLN crystal: $50 \times 2 \times 0.5 \text{ mm}^3$
4. Previous best measured output: $57 \text{ mW green, PPLN at } 59.5^\circ\text{C}$
5. Previous calculated waist: about $45 \text{ }\mu\text{m}$
6. Desired next target: about $77\text{--}80 \text{ }\mu\text{m}$

2 Definitions and Equations

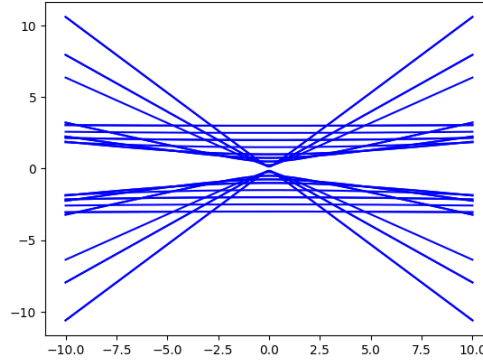
Beam Diameter ($\frac{1}{e^2}$): The $\frac{1}{e^2}$ beam diameter captures the diameter where the intensity of the beam drops to 13.53%; in other words, 86.47% of the beam intensity lies within the given diameter.



Beam Waist (ω_0): The beam waist is the portion of the beam where the radius is minimized, denoted ω_0 . The tighter the focus the smaller the waist, but this introduces a large amount of divergence, as seen in the z -dependent beam width equation

$$\omega^2(z) = \omega_0^2 \left[1 + \left(\frac{\lambda z}{\pi \omega_0^2} \right)^2 \right] \quad (1)$$

Small waists are achieved by a “short focal length and a high numerical aperture largely filled by the input beam.”[?]



Radius of Curvature (R_g): The radius of curvature is defined by Eq. ?? . At its boundaries, $\lim_{z \rightarrow 0} R_g(z) = \infty$ where the wavefront is perfectly flat and acts like a section of a circle with infinite radius, and $\lim_{z \rightarrow \infty} R_g(z) = z$ where the wavefront averages out to a radius equal to the distance from the waist.

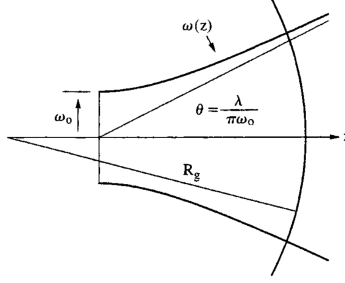
$$R_g(z) = z \left[1 + \left(\frac{\pi \omega_0^2}{\lambda z} \right)^2 \right] \quad (2)$$

Rayleigh Range (z_0): The Rayleigh range is the distance at which the cross-sectional area of the beam is no more than twice that of the beam waist, ensuring at least 50% of the beam is retained.[?]

$$z_0 = \frac{\pi \omega_0^2}{\lambda} \quad (3)$$

Beam Divergence (θ): The beam divergence is the slope of the linear component of beam expansion (the far-field expansion), measured as a half-angle dependent on the wavelength and inversely on the waist:

$$\theta = \frac{\lambda}{\pi \omega_0} \quad (4)$$



Confocal Parameter (b): The confocal parameter uses the Rayleigh range to quantify the useful depth of focus—it is “the total length of the focused region around the beam waist.” [?]

Boyd–Kleinman Focusing: $h = 1.068$ at $L/b = 2.84$ [?]; $h = 1.066$ at $L/b = 2.81$ [?]. Here h is the conversion efficiency, L is the crystal length, and $b = 2z_0$ is the confocal parameter. We work backwards from these values to find the optimal waist; see Section ??.

1

2

3 Calculations

3.1 Initial Beam Parameters

Starting from a 0.56 mm $1/e^2$ beam diameter at 1064 nm, what are the approximate Gaussian beam parameters?

Using Eq. ??:³

$$z_{01} = \frac{\pi \omega_{01}^2}{\lambda} = \frac{\pi \cdot (0.00028)^2}{1.064 \times 10^{-6}} \approx 0.23149 \text{ m} \approx 23.15 \text{ cm}$$

This results in $L/b = 0.108$, far from the Boyd–Kleinman optimum.

Using Eq. ??:

$$\theta = \frac{\lambda}{\pi \omega_{01}} = \frac{1.064 \times 10^{-6}}{\pi \cdot 0.00028} \approx 1.210 \times 10^{-3} \text{ rad}$$

3.2 Boyd–Kleinman Focusing Targets

3.2.1 Optimum: $L/b = 2.84$ [?]

Given $L/b = 2.84$ and $L = 50 \text{ mm}$.

$$b = \frac{L}{2.84} = \frac{0.05}{2.84} \text{ m}$$

Using Eq. ??:

$$\frac{0.05}{2.84} = b = 2z_{02} = \frac{2\pi \omega_{02}^2}{\lambda} \Rightarrow \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot 2.84}} \approx 54.602 \text{ } \mu\text{m}$$

3.2.2 Optimum: $L/b = 2.81$ [?]

Given $L/b = 2.81$ and $L = 50 \text{ mm}$.

$$b = \frac{L}{2.81} = \frac{0.05}{2.81} \text{ m}$$

Using Eq. ??:

$$\frac{0.05}{2.81} = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \Rightarrow \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot 2.81}} \approx 54.892 \mu\text{m}$$

3.2.3 Conservative rule of thumb: $L/b = 1$ [?]

4

Given $L/b = 1$ and $L = 50 \text{ mm}$.

$$b = L = 0.05 \text{ m}$$

Using Eq. ??:

$$0.05 = b = 2z_{02} = \frac{2\pi\omega_{02}^2}{\lambda} \Rightarrow \omega_{02} = \sqrt{\frac{1064 \times 10^{-9} \cdot 0.05}{2\pi}} \approx 92.017 \mu\text{m}$$

3.2.4 Working Backwards from Target Waists

To recover a desired ω_{02} , we solve for $\xi \equiv L/b$:

1. $\omega_{02} = 45 \mu\text{m}$:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (45 \times 10^{-6})^2} \approx 4.181$$

This ratio is too large (waist too tight), leading to excessive diffraction and reduced efficiency.

2. $\omega_{02} = 77 \mu\text{m}$:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (77 \times 10^{-6})^2} \approx 1.428$$

Within the proper range.

3. $\omega_{02} = 80 \mu\text{m}$:

$$\xi = \frac{1064 \times 10^{-9} \cdot 0.05}{2\pi \cdot (80 \times 10^{-6})^2} \approx 1.323$$

Within the proper range.

3.3 Collimator Waist Estimation

Repeating the analysis without assuming the beam waist is at the collimator face.⁵

1. Using Eq. ?? to recover the waist from the measured divergence:

$$\omega_{01} = \frac{\lambda}{\pi\theta} = \frac{1.064 \times 10^{-6}}{\pi \cdot 0.00121} \approx 279.9 \mu\text{m}$$

2. Using the beam propagation equation (Eq. ??, from Eq. 2-27 in [?]):

$$\omega^2(z) = \omega_{01}^2 \left[1 + \left(\frac{\lambda z}{\pi\omega_{01}^2} \right)^2 \right] \quad (5)$$

Solving for z :

$$\frac{\omega^2(z)}{\omega_{01}^2} - 1 = \left(\frac{\lambda z}{\pi \omega_{01}^2} \right)^2 \Rightarrow z = \pm \left(\sqrt{\frac{\omega^2(z)}{\omega_{01}^2} - 1} \right) \frac{\pi \omega_{01}^2}{\lambda}$$

Setting $\omega(z) = 0.00028$ m (the $1/e^2$ radius at the collimator aperture):

$$z \approx \pm \left(\sqrt{\frac{(0.00028)^2}{(0.0002799)^2} - 1} \right) \frac{\pi \cdot (0.0002799)^2}{1.064 \times 10^{-6}} \approx \pm 6.184 \text{ mm}$$

The waist is located approximately 6.2 mm *behind* the collimator face.⁶

3. Recalculating the Rayleigh range with the corrected waist:

$$z_{01} = \frac{\pi \omega_{01}^2}{\lambda} = \frac{\pi \cdot (2.799 \times 10^{-4})^2}{1.064 \times 10^{-6}} \approx 23.132 \text{ cm}$$

4. Because the beam waist $\omega_{01} \approx 280 \text{ } \mu\text{m}$ is the *minimum* radius of the laser, the beam can only diverge from that point and would “scrape” the inside of the PPLN crystal. A focusing lens is therefore required.

3.4 Lens Selection

1. The magnification relation from [?]:

$$\omega_{02} = \alpha \cdot \omega_{01} \Rightarrow \alpha = \frac{\omega_{02}}{\omega_{01}} \quad (6)$$

2. The lens-placement formula from [?]:

$$\alpha = \frac{f}{\sqrt{(|d_1| - f)^2 + z_{01}^2}} \quad (7)$$

Solving for $|d_1|$:

$$\alpha^2 [(|d_1| - f)^2 + z_{01}^2] = f^2 \Rightarrow |d_1| = \sqrt{\left(\frac{f}{\alpha} \right)^2 - z_{01}^2} + f$$

3. Image-distance formula from [?]:

$$d_2 = f + \alpha^2 (d_1 - f) \quad (8)$$

4. **Numerical values.** With $\omega_{02} = 80 \text{ } \mu\text{m}$ and $\omega_{01} = 279.9 \text{ } \mu\text{m}$:

$$\alpha = \frac{80}{279.9} \approx 0.28582, \quad z_{01} \approx 0.23132 \text{ m}$$

For a focal length f chosen in the lab:

$$d_1 = \sqrt{\left(\frac{f}{0.28582} \right)^2 - (0.23132)^2} + f, \quad d_2 = f + (0.28582)^2 (d_1 - f)$$

5. **Derivation of equations used above** Starting from [?]:

$$\frac{1}{q_1} = \frac{1}{R_g} - i \frac{\lambda}{\pi \omega_{01}^2} \quad (9)$$

where R_g is the radius of curvature of the input beam. Taking the limit $R_g \rightarrow \infty$ (collimated input):

$$q_1 = i \frac{\pi \omega_{01}^2}{\lambda} = iz_{01}$$

The ABCD ray-transfer matrix for free-propagation d_1 , thin lens f , then free-propagation d_2 is:

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & d_2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{pmatrix} \begin{pmatrix} 1 & d_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{d_2}{f} & d_1 + d_2 - \frac{d_1 d_2}{f} \\ -\frac{1}{f} & 1 - \frac{d_1}{f} \end{pmatrix}$$

so $A = 1 - \frac{d_2}{f}$, $B = d_1 + d_2 - \frac{d_1 d_2}{f}$, $C = -\frac{1}{f}$, $D = 1 - \frac{d_1}{f}$.

The q -parameter transforms as (Eq. ?? from [?]):

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D} \quad (10)$$

Substituting $q_1 = iz_{01}$:

$$q_2 = \frac{\left(1 - \frac{d_2}{f}\right) iz_{01} + \left(d_1 + d_2 - \frac{d_1 d_2}{f}\right)}{\left(-\frac{1}{f}\right) iz_{01} + \left(1 - \frac{d_1}{f}\right)}$$

Multiplying numerator and denominator by the complex conjugate of the denominator, $\left(1 - \frac{d_1}{f}\right) + i\left(\frac{z_{01}}{f}\right)$:

Real part:

$$\begin{aligned} \text{Re}(q_2) &= \frac{-\left(1 - \frac{d_2}{f}\right) \left(\frac{z_{01}^2}{f}\right) + \left(d_1 + d_2 - \frac{d_1 d_2}{f}\right) \left(1 - \frac{d_1}{f}\right)}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \\ &= \frac{d_2 \left[\left(1 - \frac{d_1}{f}\right)^2 + \frac{z_{01}^2}{f^2}\right] + d_1 \left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \\ &= d_2 + \frac{d_1 \left(1 - \frac{d_1}{f}\right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} \end{aligned}$$

Imaginary part:

$$\text{Im}(q_2) = \frac{z_{01} \left[\left(1 - \frac{d_1}{f} - \frac{d_2}{f} + \frac{d_1 d_2}{f^2}\right) + \left(\frac{d_1}{f} + \frac{d_2}{f} - \frac{d_1 d_2}{f^2}\right)\right]}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2} = \frac{z_{01}}{\left(\frac{z_{01}}{f}\right)^2 + \left(1 - \frac{d_1}{f}\right)^2}$$

Full expression:

$$q_2 = \left[d_2 + \frac{d_1 \left(1 - \frac{d_1}{f} \right) - \frac{z_{01}^2}{f}}{\left(\frac{z_{01}}{f} \right)^2 + \left(1 - \frac{d_1}{f} \right)^2} \right] + i \left[\frac{z_{01}}{\left(\frac{z_{01}}{f} \right)^2 + \left(1 - \frac{d_1}{f} \right)^2} \right]$$

Setting the real part to zero to satisfy the waist condition: $\lim_{z \rightarrow 0} \frac{1}{R_g(z)} = 0$ results in

$$d_2 = \frac{f[(d_1 - f)^2 + z_{01}^2] + f^2(d_1 - f)}{(d_1 - f)^2 + z_{01}^2} = f + \frac{f^2(d_1 - f)}{(d_1 - f)^2 + z_{01}^2}$$

$$d_2 = f + \alpha^2(d_1 - f)$$

proving Eq. ??, where $\alpha^2 = \frac{f^2}{(d_1 - f)^2 + z_{01}^2}$ follows Eq. ??.

4 Use Case

PPLN:⁷ To get the most out of our PPLN crystals, four key aspects must be considered:

1. Crystal Length

- (a) Decided by available materials, manufacturing capabilities and cost.⁸

2. Polarization

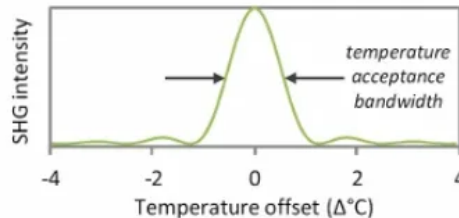
- (a) The input light must be e-polarized, i.e. the polarization must be aligned with the dipole moment of the crystal. This is accomplished by aligning the polarization axis of the light parallel to the thickness of the crystal.[?]

3. Focusing and the Optical Arrangement

- (a) For SHG with CW lasers, the Boyd–Kleinman result shows that optimum conversion efficiency is achieved when $L/b = 2.84$.[?]

4. Temperature and Period

- (a) The crystal is manufactured with periodic stripes of alternating polarity with a spacing called the poling period. This quasi-phase-matched arrangement allows the harmonic power to build up continuously.
- (b) The effective poling period can be tuned by changing the crystal temperature, which causes thermal expansion or contraction.
- (c) Note: The refractive index of the crystal also changes with temperature.
- (d) The output intensity follows a sinc^2 curve (shown below), defining the temperature acceptance bandwidth.



- (e) Acceptance bandwidth is inversely proportional to crystal length.
- (f) The optimum temperature can be found by heating the crystal to 20°C above the calculated phase-matching temperature, then allowing it to cool while monitoring the generated output power.[?]

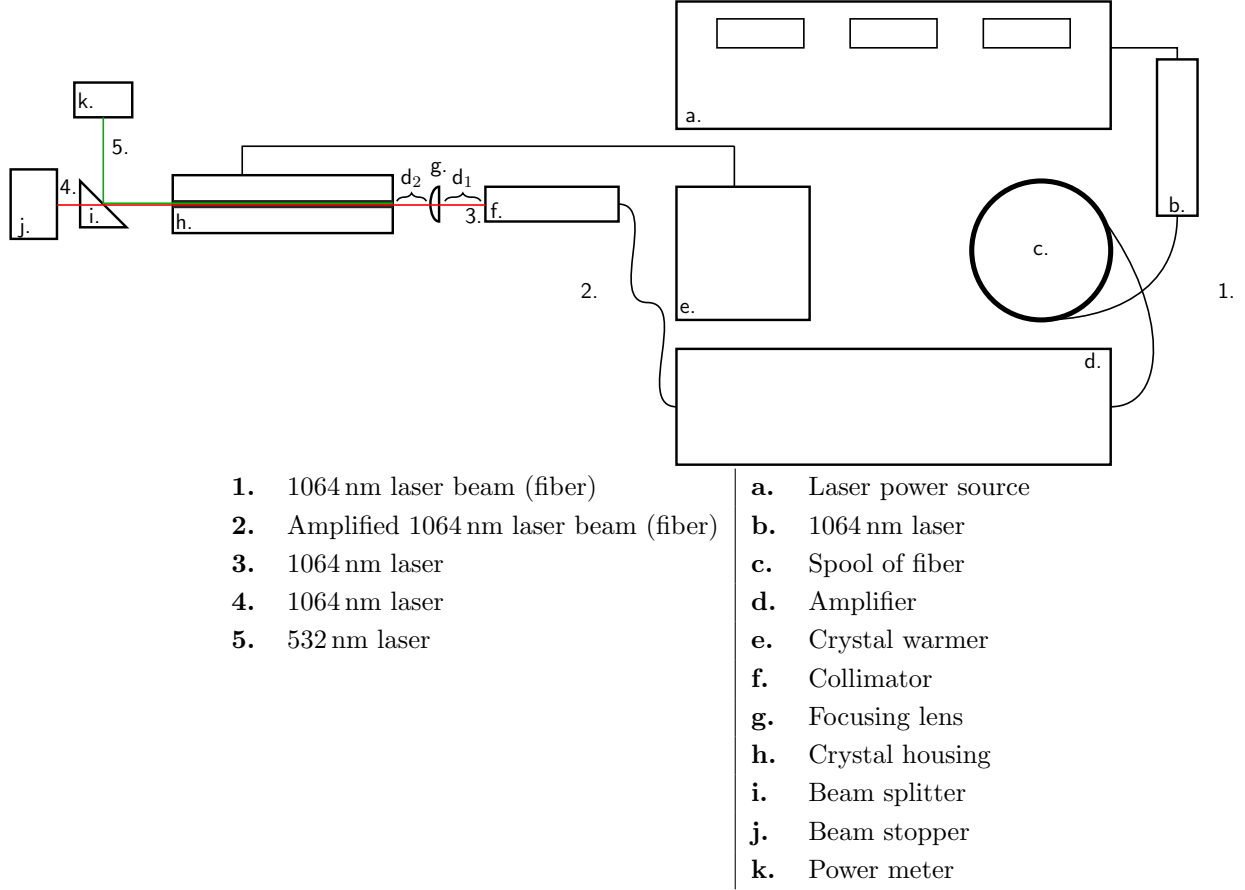


Figure 1: Experimental setup and component legend.

5 Experimental Results

5.1 Output Power vs. Crystal Temperature

add graph

5.2 Achieved Waist Measurement

Add measurements

5.3 Comparison with Boyd–Kleinman Prediction

interms of efficacy check inbetween 2.81 and 1

6 Error Analysis

- Uncertainty in ω_{01} from divergence measurement: $\pm X\%$
- Tolerance on lens placement d_1 : ± 1 mm leads to waist change of ...
- Total uncertainty in achieved ω_{02} : $\pm Y \mu\text{m}$

7 Operating Instructions

7.1 Prerequisites/Safety

- Ensure curtains are closed and laser indicator is on outside the lab.
- Take off all jewelry and don appropriate laser safety goggles.
- Verify the beam path is clear of obstructions and that all optical components are securely mounted.
- Verify the power meter is aligned and is properly rated for the expected power levels, default to 500 mW.
- Initiate appropriate music... a good fall back is Led Zeppelin, Black Sabbath, or AC/DC.

7.2 Diode Initialization

1. Turn on the diode, it is located on the top shelf on the far right, it is blue.
2. Check the current and temperature settings on the diode controller, Temp should be 19.5°C and Current should be 165 mA.
3. Turn on the temperature via TEC controller, push ON.
4. Turn on the current via the controller, push ON.

7.3 PPLN Initialization

1. Verify HCP is set to 59.5°C, then turn on the PPLN temperature controller.

PUT ON SAFETY GOGGLES BEFORE PROCEEDING FURTHER

7.4 Amplifier Initialization

1. Turn on the amplifier, but leave the key switched off for now.
2. Verify the amplifier settings: the current should be set to 20 mA.
3. Turn on the amplifier key switch, then wait for system to boot through serial number and loading screens.
4. Turn on the amplifier output by turning the dial to highlight *START*, then push the dial in.

7.5 Beam Alignment

1. Verify collimator output is centered on iris.
2. Place focusing lens at distance d_1 (see Sec. ??).
3. ...
4. Use IR viewer/detector card to check that beam is not clipped by the crystal housing.

7.6 How to Shut Down the Laser System

1. Turn the key to off on the amplifier and then shut it down.
2. Flip the current off and then the TEC peltier device off.
3. Turn off the crystal heater

7.7 Troubleshooting

Placeholder for troubleshooting instructions.

8 Optical Lab Components Inventory

Part Num	Specifications	Description
Thor Labs Bin #1: Lenses		
LA1951	$f = 25.4$ mm	PCX $\varnothing 25.4$ mm
LA1805	$f = 30$ mm	PCX $\varnothing 25.4$ mm
LA1027	$f = 35$ mm	PCX $\varnothing 25.4$ mm
LA1422	$f = 40$ mm	PCX $\varnothing 25.4$ mm
LA1131	$f = 50$ mm	PCX $\varnothing 25.4$ mm
LA1134	$f = 60$ mm	PCX $\varnothing 25.4$ mm
LA1608	$f = 75$ mm	PCX $\varnothing 25.4$ mm
LA1509	$f = 100$ mm	PCX $\varnothing 25.4$ mm
LA1986	$f = 125$ mm	PCX $\varnothing 25.4$ mm
LA1433	$f = 150$ mm	PCX $\varnothing 25.4$ mm
LA1229	$f = 175$ mm	PCX $\varnothing 25.4$ mm
LA1708	$f = 200$ mm	PCX $\varnothing 25.4$ mm
LA1461	$f = 250$ mm	PCX $\varnothing 25.4$ mm
LA1484	$f = 300$ mm	PCX $\varnothing 25.4$ mm
LA1172	$f = 400$ mm	PCX $\varnothing 25.4$ mm
LA1908	$f = 500$ mm	PCX $\varnothing 25.4$ mm
LA1978	$f = 750$ mm	PCX $\varnothing 25.4$ mm
LA1464	$f = 1000$ mm	PCX $\varnothing 25.4$ mm
LB1761	$f = 25.4$ mm	BCX $\varnothing 25.4$ mm
LB1757	$f = 30$ mm	BCX $\varnothing 25.4$ mm
LB1811	$f = 35$ mm	BCX $\varnothing 25.4$ mm
LB1027	$f = 40$ mm	BCX $\varnothing 25.4$ mm
LB1471	$f = 50$ mm	BCX $\varnothing 25.4$ mm
LB1596	$f = 60$ mm	BCX $\varnothing 25.4$ mm
LB1901	$f = 75$ mm	BCX $\varnothing 25.4$ mm
LB1676	$f = 100$ mm	BCX $\varnothing 25.4$ mm
LB1904	$f = 125$ mm	BCX $\varnothing 25.4$ mm
LB1437	$f = 150$ mm	BCX $\varnothing 25.4$ mm
LB1294	$f = 175$ mm	BCX $\varnothing 25.4$ mm
LB1945	$f = 200$ mm	BCX $\varnothing 25.4$ mm
LB1056	$f = 250$ mm	BCX $\varnothing 25.4$ mm
LB1779	$f = 300$ mm	BCX $\varnothing 25.4$ mm
LB1391	$f = 400$ mm	BCX $\varnothing 25.4$ mm
LB1869	$f = 500$ mm	BCX $\varnothing 25.4$ mm
LB1475	$f = 750$ mm	BCX $\varnothing 25.4$ mm
LB1409	$f = 1000$ mm	BCX $\varnothing 25.4$ mm
LC1715	$f = -50$ mm	PCV $\varnothing 25.4$ mm
LC1582	$f = -75$ mm	PCV $\varnothing 25.4$ mm
LC1120	$f = -100$ mm	PCV $\varnothing 25.4$ mm
LD1464	$f = -50$ mm	BCV $\varnothing 25.4$ mm
LD1170	$f = -75$ mm	BCV $\varnothing 25.4$ mm

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Table 1 – continued from previous page

Part Num	Specifications	Description
LD1613	$f = -100$ mm	BCV $\varnothing 25.4$ mm
LE	—	MP $\varnothing 25.4$ mm
LE1156	$f = 125$ mm	MP $\varnothing 25.4$ mm
LE1104	$f = 150$ mm	MP $\varnothing 25.4$ mm
LE1202	$f = 200$ mm	MP $\varnothing 25.4$ mm
LE1157	$f = 250$ mm	MP $\varnothing 25.4$ mm
LE1929	$f = 300$ mm	MP $\varnothing 25.4$ mm
LE1872	$f = 400$ mm	MP $\varnothing 25.4$ mm
LE1261	$f = 500$ mm	MP $\varnothing 25.4$ mm
LE1458	$f = 1000$ mm	MP $\varnothing 25.4$ mm
LF1822	$f = -100$ mm	MN $\varnothing 25.4$ mm
LF1510	$f = -125$ mm	MN $\varnothing 25.4$ mm
LF1547	$f = -150$ mm	MN $\varnothing 25.4$ mm
LF1097	$f = -200$ mm	MN $\varnothing 25.4$ mm
LF1774	$f = -250$ mm	MN $\varnothing 25.4$ mm
LF1015	$f = -300$ mm	MN $\varnothing 25.4$ mm
LF1544	$f = -400$ mm	MN $\varnothing 25.4$ mm
LF1988	$f = -500$ mm	MN $\varnothing 25.4$ mm
LF1141	$f = -1000$ mm	MN $\varnothing 25.4$ mm
Thor Labs Bin #2: Mounting Bases & Post Holders		
PH1	1" Long	$\varnothing 1/2''$ Post Holder
PH1.5	1.5" Long	$\varnothing 1/2''$ Post Holder
PH2	2" Long	$\varnothing 1/2''$ Post Holder
PH3	3" Long	$\varnothing 1/2''$ Post Holder
PH4	4" Long	$\varnothing 1/2''$ Post Holder
PH6	6" Long	$\varnothing 1/2''$ Post Holder
CL5	—	Standard Base Clamp
BA1S	$1'' \times 2.3'' \times 3/8''$	Mounting Base
BA1	$1'' \times 3'' \times 3/8''$	Mounting Base
BA2	$2'' \times 3'' \times 3/8''$	Mounting Base
Thor Labs Bin #3: Posts, Clamps, & Accessories		
TR075	$L = 0.75''$	$\varnothing 1/2''$ Stainless Steel Posts
TR1	$L = 1''$	$\varnothing 1/2''$ Stainless Steel Posts
TRK05	Alignment Key	$\varnothing 1/2''$ Stainless Steel Posts
TR1.5	$L = 1.5''$	$\varnothing 1/2''$ Stainless Steel Posts
TR2	$L = 2''$	$\varnothing 1/2''$ Stainless Steel Posts
TR3	$L = 3''$	$\varnothing 1/2''$ Stainless Steel Posts
TR4	$L = 4''$	$\varnothing 1/2''$ Stainless Steel Posts
TR6	$L = 6''$	$\varnothing 1/2''$ Stainless Steel Posts
TR8	$L = 8''$	$\varnothing 1/2''$ Stainless Steel Posts
TRA2	$L = 2''$	$\varnothing 1/2''$ Aluminum Posts
TRA3	$L = 3''$	$\varnothing 1/2''$ Aluminum Posts
R2	0.44" Thick	Slip-On Collars for $\varnothing 1/2''$ Posts

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Table 1 – continued from previous page

Part Num	Specifications	Description
R2T	0.23" Thick	Slip-On Collars for $\varnothing 1/2''$ Posts
PMTR	Component Clamp	Slip-On Collars for $\varnothing 1/2''$ Posts
TR1M	1 mm Thick	Spacers for $\varnothing 1/2''$ Posts
TR3M	3 mm Thick	Spacers for $\varnothing 1/2''$ Posts
TR5M	5 mm Thick	Spacers for $\varnothing 1/2''$ Posts
TR10M	10 mm Thick	Spacers for $\varnothing 1/2''$ Posts
RA90	Right-Angle Clamp	Clamps and Mounts for $\varnothing 1/2''$ Posts
RA180	Right-Angle End Clamp	Clamps and Mounts for $\varnothing 1/2''$ Posts
SWC	Swivel Clamp	Clamps and Mounts for $\varnothing 1/2''$ Posts
TRB1	Ball and Socket Mount (8-32)	Clamps and Mounts for $\varnothing 1/2''$ Posts
Thor Labs Bin #4: SM1 Lens Tubes & Optomechanics		
SM1L03	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L05	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L10	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L15	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L20	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L30	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1L40	—	SM1 Lens Tubes for $\varnothing 1''$ Optics
SM1E60	—	Extension Tube for $\varnothing 1''$ Optics
SM1M05	—	SM1 Lens Tubes Without External Threads
SM1M10	—	SM1 Lens Tubes Without External Threads
SM1M20	—	SM1 Lens Tubes Without External Threads
SM1M30	—	SM1 Lens Tubes Without External Threads
SM1S10	—	SM1 Lens Tube Spacers
SM1S20	—	SM1 Lens Tube Spacers
SM1L20C	—	SM1 Slotted Lens Tubes
SM1L30C	—	SM1 Slotted Lens Tubes
SM1V05	—	Adjustable SM1 Lens Tubes
SM1V10	—	Adjustable SM1 Lens Tubes
SM1V15	—	Adjustable SM1 Lens Tubes
SM1ZM	—	SM1 Zoom Housings
??	—	SM1 Zoom Housings
AD9.5F	—	SM1-Threaded Adapters for Cylindrical Components
AD11F	—	SM1-Threaded Adapters for Cylindrical Components
AD12F	—	SM1-Threaded Adapters for Cylindrical Components
AD15F	—	SM1-Threaded Adapters for Cylindrical Components
SM1SH1	—	SM1 Beam Shutter
SM1D12D	—	SM1 Iris Diaphragm
SM1B3	—	SM1 Lens Tube Bellows

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Table 1 – continued from previous page

Part Num	Specifications	Description
SM1TC	—	Clamp for SM1 Lens Tubes
SM1PL	—	SM1 Plug
SM1CP2	—	SM1 End Caps
SM1CP1	—	SM1 End Caps
VRCSM1	—	Alignment Disks
VRC2SM1	—	Alignment Disks
VRC4SM1	—	Alignment Disks
SM1T1	—	SM1 Lens Tube Couplers
SM1T2	—	SM1 Lens Tube Couplers
SM1T10	—	SM1 Lens Tube Couplers
SM1NT1	—	SM1 Slotted Locking Ring
SPW502	—	Spanner Wrenches
SPW602	—	Spanner Wrenches
SM1RR	—	Retaining Rings
SM1LTRR	—	Retaining Rings
Newport MKS Bin		
LH-0.5A	Ø0.5", 6 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-50R25	Ø0.25" Adapter, 3.5 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-1A	Ø1", 7 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-f.5A	1.5", 9 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-30M	Ø30 mm, 7 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-2A	Ø2", 9 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-3A	Ø3", 9 mm Max Edge	A-Line Lens Mounts, 8-32 Thread
LH-05N	Ø0.5", 7 mm Max Edge	A-Line Lens Mounts, Insertable
LH-1N	Ø1", 10 mm Max Edge	A-Line Lens Mounts, Insertable
LH-2N	Ø2", 13 mm Max Edge	A-Line Lens Mounts, Insertable
LH1-ATP	Alignment Target	A-Line Accessories, 8-32 Thread
LH-K1	Ø0.5" to Ø1" Link Block	A-Line Accessories, 8-32 Thread
LH-K2	Ø1" to Ø2" Link Block	A-Line Accessories, 8-32 Thread
LH-GR-4	$L = 4''$	A-Line Guide Rods
LH-GR-8	$L = 8''$	A-Line Guide Rods
LH-GR-10	$L = 10''$	A-Line Guide Rods
LFTO-WH	Ø1" Retaining Ring	Spanner Wrenches
SW-OM	Ø0.25" to Ø2" Retaining Ring	Spanner Wrenches
Edmund Industrial Optics		
DCX 12	Ø1/2"	Bi-Convex Lens
DCX 24	Ø1/2"	Bi-Convex Lens
DCX 36	Ø1/2"	Bi-Convex Lens
DCX 48	Ø1/2"	Bi-Convex Lens
DCX 60	Ø1/2"	Bi-Convex Lens
PCX 12	Ø1/2"	Plano-Convex Lens
PCX 24	Ø1/2"	Plano-Convex Lens

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Table 1 – continued from previous page

Part Num	Specifications	Description
PCX 36	$\varnothing 1/2''$	Plano-Convex Lens
PCX 48	$\varnothing 1/2''$	Plano-Convex Lens
PCX 60	$\varnothing 1/2''$	Plano-Convex Lens
PCV 12	$\varnothing 1/2''$	Plano-Concave Lens
PCV 24	$\varnothing 1/2''$	Plano-Concave Lens
PCV 48	$\varnothing 1/2''$	Plano-Concave Lens
DCV 12	$\varnothing 1/2''$	Bi-Concave Lens
EFL 12	$\varnothing 1/2''$	Lens (EFL = 12)
EFL 15	$\varnothing 1/2''$	Lens (EFL = 15)
EFL 18	$\varnothing 1/2''$	Lens (EFL = 18)
EFL 24	$\varnothing 1/2''$	Lens (EFL = 24)
EFL 30	$\varnothing 1/2''$	Lens (EFL = 30)
EFL 36	$\varnothing 1/2''$	Lens (EFL = 36)
EFL 42	$\varnothing 1/2''$	Lens (EFL = 42)
EFL 48	$\varnothing 1/2''$	Lens (EFL = 48)
EFL 54	$\varnothing 1/2''$	Lens (EFL = 54)
EFL 60	$\varnothing 1/2''$	Lens (EFL = 60)
EFL 72	$\varnothing 1/2''$	Lens (EFL = 72)
EFL 84	$\varnothing 1/2''$	Lens (EFL = 84)
EFL 100	$\varnothing 1/2''$	Lens (EFL = 100)
MKS Newport 8-32 Hardware Kit (Item SK-1)		
N8S	#8	Nut
W8S038	#8	Washer
SH8S025	$8 - 32 \times 1/4''$	Cap Screw
SH8S038	$8 - 32 \times 3/8''$	Cap Screw
SH8S050	$8 - 32 \times 1/2''$	Cap Screw
SH8S075	$8 - 32 \times 3/4''$	Cap Screw
SH8S100	$8 - 32 \times 1''$	Cap Screw
SH8S125	$8 - 32 \times 1 1/4''$	Cap Screw
SH8S150	$8 - 32 \times 1 1/2''$	Cap Screw
General Lab Instruments & Accessories		
KPX058	$f = 88.3 \text{ mm}, \varnothing 12.7 \text{ mm}$	PCX Lens, BK7 uncoated (Newport)
TC4	Hex/Allen Keys	Thorlabs Tool Kit
LB1	$400 \text{ nm} - 2 \mu\text{m}, 10\text{W}$	Beam Block (Thorlabs)
BS040	$400 - 700 \text{ nm}, 12.7 \text{ mm}$	10:90 Non-Pol Beamsplitter Cube
ULM-TILT	—	Collimator Holder (Newport)
MT-RS	—	Optical Rotation Mount for Polarizer
TC-038D	SC2502260152-02	HCP Crystal Oven Controller
PM5020	—	Dual Channel Optical Power/Energy Meter
LDC-3726	—	ILX Lightwave Laser Diode Controller
TFN-1064.502	—	TeraXion Narrowband Tunable Optical Filter
LDM-4980	—	ILX Lightwave Laser Diode Mount

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Part Num	Specifications	Description
ULS2048XL-EVO	—	Avantes Sensline Spectrometer
Keopsys	—	Lumibird Laser
VRC2	400–640 nm, 800–1700 nm	Detector Card (Thorlabs)
VRC5	700–1400 nm	Detector Card (Thorlabs)
PM-1064-10	—	Optilab RF Phase Modulator
SC2502260152-02	P:3.2 I:33 D:6	Crystal Oven
UP-1A	—	Newport Universal Post
S-2000 Series	—	Stabilizer Laminar Flow Isolator (Newport)
RS2000	—	Sealed Hole Table Top (Newport)
ATS Series	—	Overhead Table Shelf System (Newport)
Lens Tissues	—	Thorlabs Lens Cleaning Tissues
Statshield	—	ESD Garment
BB-M	—	Broadband Mirror (Newport)
HW-KIT6	—	Thorlabs Thread Adapter Kit
HW-KIT2	1/4"-20	Thorlabs Hardware Kit

Notes

¹Wikipedia entry: Second-harmonic generation (SHG), also known as frequency doubling, is the lowest-order wave-wave nonlinear interaction that occurs in various systems, including optical, radio, atmospheric, and magnetohydrodynamic systems. As a prototype behavior of waves, SHG is widely used, for example, in doubling laser frequencies. SHG was initially discovered as a nonlinear optical process in which two photons with the same frequency interact with a nonlinear material, are “combined”, and generate a new photon with twice the energy of the initial photons (equivalently, twice the frequency and half the wavelength), conserving the coherence of the excitation. It is a special case of two-photon sum-frequency generation, and more generally of harmonic generation.

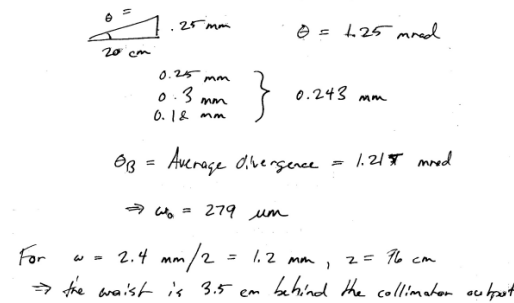
²**Phase-matching:** For sum-frequency generation to occur efficiently, phase-matching conditions must be satisfied: $\hbar k_3 \approx \hbar k_1 + \hbar k_2$, where k_1, k_2, k_3 are the angular wavenumbers of the three waves travelling through the medium. (This equation resembles conservation of momentum.) The sum-frequency generation becomes more efficient as this condition is satisfied more precisely.

³In this calculation I assume the full beam diameter $2\omega_{01} = 0.56$ mm because the beam emerges from a collimator. This assumes the beam waist may be located a short distance away from the collimator face, as noted in [?]: “The purpose of a beam collimator is essentially to transform a strongly diverging light beam into a collimated beam, i.e., a beam where light propagates essentially only in one direction, and the beam divergence is weak. The output beam may have its beam waist close to the output aperture, or a mild focus somewhat away from it.”

⁴“In general, a good rule of thumb is that the spot size should be chosen such that the Rayleigh range is half the length of the crystal.” If $z_{02} = L/2$, then $b = 2z_{02} = L$, so $L/b = 1$.

⁵Note: 1.21 mRad was a measured value, substantially lower than the spec sheet value of approximately 5 mRad.

⁶The following calculations were made under the (incorrect) assumption of a waist radius of $45 \mu\text{m}$ rather than the updated $279.9 \mu\text{m}$ derived here.



Handwritten calculations showing beam divergence and waist radius:

$$\theta = \frac{0.25 \text{ mm}}{20 \text{ cm}} \quad \theta = 1.25 \text{ mrad}$$

$$\left. \begin{array}{l} 0.25 \text{ mm} \\ 0.3 \text{ mm} \\ 0.12 \text{ mm} \end{array} \right\} 0.243 \text{ mm}$$

$$\theta_B = \text{Average divergence} = 1.21 \text{ mrad}$$

$$\Rightarrow \omega_0 = 279 \mu\text{m}$$

For $w = 2.4 \text{ mm}/2 = 1.2 \text{ mm}$, $z = 76 \text{ cm}$
 $\Rightarrow$ the waist is 3.5 cm behind the collimator output

⁷Periodically poled lithium niobate (PPLN) is a lithium niobate crystal that switches polarity in a periodic fashion and is used for achieving quasi-phase-matching in nonlinear optics.

⁸Each crystal has an associated pump acceptance bandwidth $\Delta\lambda \approx \frac{0.44\lambda}{L|n_z'(\lambda) - 0.5n_z'(\lambda/2)|}$ which is inversely dependent on length, so crystal band SHG pulse.